

SQL Where vs Having

<https://www.essentialsql.com/what-is-the-difference-between-where-and-having-clauses/>

- A WHERE clause is used to filter records from a result. The filter occurs before any groupings are made.
- A HAVING clause is used to filter values from a group.

Before we go any further let's review the format of an SQL Statement. It is

```
SELECT
FROM
WHERE
GROUP BY
HAVING
```

```
SELECT    SalesOrderID,
          SUM(UnitPrice * OrderQty) AS TotalPrice
FROM      Sales.SalesOrderDetail
GROUP BY  SalesOrderID
HAVING    SalesOrderID > 50000
```

```
SELECT    SalesOrderID,
          SUM(UnitPrice * OrderQty) AS TotalPrice
FROM      Sales.SalesOrderDetail
GROUP BY  SalesOrderID
HAVING    SUM(UnitPrice * OrderQty) > 10000
          AND SalesOrderID > 50000
```

```
SELECT    SalesOrderID,
          SUM(UnitPrice * OrderQty) AS TotalPrice
FROM      Sales.SalesOrderDetail
WHERE      SalesOrderID > 50000
GROUP BY  SalesOrderID
HAVING    SUM(UnitPrice * OrderQty) > 10000
```

```
SELECT    SalesOrderID,
          SUM(UnitPrice * OrderQty) AS TotalPrice
FROM      Sales.SalesOrderDetail
GROUP BY  SalesOrderID
HAVING    SUM(UnitPrice * OrderQty) > 10000 AND LineTotal > 10
```

Actually that query generates an error. The column LineTotal is not part of the group by field list nor the result of an aggregate total.

```
SELECT    SalesOrderID,  
          SUM(UnitPrice * OrderQty) AS TotalPrice  
FROM      Sales.SalesOrderDetail  
WHERE     LineTotal > 100  
GROUP BY SalesOrderID  
HAVING    SUM(UnitPrice * OrderQty) > 10000
```

To summarize the difference between WHERE and HAVING:

- WHERE is used to filter records before any groupings take place.
- HAVING is used to filter values after they have been grouped. Only columns or expressions in the group can be included in the HAVING clause's conditions..